# THABANG WASHINGTON MANYAMA

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## PROFILE

Thabang Manyama, 21. Final-year BSc Data Science student who builds and ships production-grade analytical work, not just coursework. Author of an open-source ML observability library validated on national labour-force microdata. Led the analysis, modelling and deployment of a hung-council election forecasting dashboard for the DIRISA Student Datathon Challenge, and now leads Ubufakazi, a live digital forensics tool built for HackSecure 2026 that turns scam victims' chats and payment slips into tamper-evident case files. Comfortable end to end: sourcing and validating raw institutional data, building the model, testing it properly, and shipping a dashboard someone can actually use. 14 module distinctions to date, 2× Academic Top Achiever at EDUVOS (2025 and 2026), Golden Key member, and 3rd place at the ITWeb Security Summit Hackathon.

## TECHNICAL SKILLS

**Languages** Python, R, SQL, JavaScript
**Data & ML** pandas, NumPy, scikit-learn, XGBoost, statsmodels, SciPy, Jupyter. Supervised and unsupervised learning, feature engineering, model evaluation and monitoring
**Statistics** Hypothesis and permutation testing, OLS, distributed-lag and fixed-effects regression, correlation analysis, descriptive and inferential statistics, linear algebra
**Visualisation** Power BI, Tableau, Streamlit, Matplotlib, D3.js, Excel
**Engineering** Git and GitHub, GitHub Actions CI, pytest, Playwright, package distribution, Netlify deployment, relational databases and SQL query design, object-oriented design
**Security & Cloud** Digital forensics (SHA-256 evidence hashing, HMAC-keyed identifiers), Content Security Policy, session monitoring, AI-driven risk analysis, network security, Azure, Microsoft Fabric, Kubernetes and cloud-native fundamentals

## SELECTED PROJECTS

**Hung Council Watch - 2026 Local Election Forecasting** 2026

*DIRISA Student Datathon Challenge (CSIR) | Election forecasting, ML and deployment · eduvos-hung-councils.netlify.app · github.com/Don4chnge/Eduvos-Team-DIRISA-2026*

- **Forecast which of South Africa's 213 municipalities are most likely to have a hung council** after the 2026 Local Government Elections, using official IEC results from five elections (2000 to 2021).
- **Built the exploratory analysis and data pipeline:** linked 263 councils across boundary changes, engineered fragmentation (Effective Number of Parties), margin and turnout features, and uncovered two source-data errors: a 2016 turnout column that counted only the leading party's votes, and 2021 data truncated by Excel's row limit.
- **Tested and rejected the original inverted-U turnout hypothesis** with Lind-Mehlum U-tests and within-municipality fixed-effects models, and used the evidence to redirect the project to a stronger research question.
- **Built and validated the ML pipeline:** KMeans competition regimes and four forecasting models (baseline, two logistic regressions, tuned XGBoost) compared with grouped and time-ordered validation; the selected model flagged 62 of 72 hung councils on an unseen election (ROC-AUC 0.88) and forecasts about 77 hung councils in 2026.
- **Designed and deployed a public interactive dashboard** (D3.js, Netlify) with a risk map, swing scenarios and an election-day simulator, running the trained model directly in the browser.
- **Made the full pipeline reproducible from raw data,** with relative paths, a code-based rebuild of the 2021 data and a documented GitHub repository, and produced the project slides and technical documentation.

**Valorian Intelligence - ML Observability Toolkit** 2026

*Open-source Python library | Drift detection and model health monitoring · github.com/Don4chnge/valorian-intelligence*

- **Built an open-source Python library** that detects data and concept drift in deployed ML models using PSI, Kolmogorov-Smirnov and chi-square tests, plus a domain-classifier approach that catches multivariate shifts per-feature tests miss.
- **Designed DriftScore,** a composite 0-100 model health metric weighting performance degradation, worst-feature PSI and drift breadth, with documented rationale behind every weight.
- **Validated on six quarters of Stats SA Quarterly Labour Force Survey microdata** (~40,000 respondents per quarter): zero false alarms across five real batches, and a multivariate distribution shift confirmed by permutation test ($p = 0.048$) that univariate monitoring never registered.
- **Shipped with a 78-test suite** covering detection accuracy and false-positive resistance, CI across Python 3.10 to 3.12, and a deployed Streamlit dashboard.

**Ubufakazi - Scam Evidence and Victim-Linking Platform** 2026

*HackSecure 2026 (CyberM8), Team CyberThreat lead | Digital forensics · ubufakazi.netlify.app · github.com/Don4chnge/ubufakazi*

- **Led a three-person team to build and ship a live digital forensics tool** that turns a scam victim's WhatsApp chats and proof-of-payment files into a tamper-evident, police-ready case file in under 60 seconds.
- **Implemented evidence locking with SHA-256** (Web Crypto API): every file and the chat text are fingerprinted on upload, with a case-manifest hash so any later edit, missing or swapped file is detectable.
- **Built a multilingual extraction engine** that parses Android and iPhone WhatsApp exports, pulls South African bank accounts, phone numbers, links and rand amounts, and rebuilds the scam timeline in English, isiZulu, isiXhosa, Afrikaans and Sesotho.
- **Designed a privacy-preserving "Scam Fingerprint"** that links victims through keyed HMAC-SHA256 hashes and message-script similarity, so a syndicate stays linked even after it changes phone numbers and bank accounts.

- **Validated on a 70-case, five-language test pack** with an automated headless-browser test (every case extracts the correct account, amount and scam type), and deployed on Netlify with a strict Content Security Policy.

**Secure Session Guardian** 2026

*Cybersecurity platform | Session integrity and account takeover defence*

- **Built a platform protecting users** from session hijacking, token theft and account takeover attacks.
- **Implemented continuous session monitoring with AI-driven risk analysis** to detect anomalous behaviour and credential misuse in real time.
- **Designed adaptive security controls** that respond dynamically to threat level, balancing user experience against strength of protection.
- **Placed 3rd at the ITWeb Security Summit Hackathon** for innovation, technical execution and security impact.

**SmartRetail Demand Forecasting System** 2025

*Machine Learning Algorithms module (ITMLA2-B44)*

- **Developed a retail demand forecasting system** using machine learning on sales, seasonal and customer data.
- **Built Python data pipelines** for cleaning and analysis, plus dashboards and performance monitoring to support inventory decisions and track model accuracy.

## HONOURS & AWARDS

- **3rd Place, ITWeb Security Summit Hackathon (2026).** Recognised for Secure Session Guardian, a platform defending against session hijacking and account takeover.
- **2× Academic Top Achiever, EDUVOS (2025 and 2026).** Top-performing student, BSc Information Technology: Data Science, in two consecutive years.
- **Golden Key International Honour Society, Member (2025).** Invited membership, awarded to students in the top 15% of their academic year.

## EDUCATION

**BSc Information Technology: Data Science** Feb 2024 - Present

*EDUVOS | NQF Level 7, SAQA ID 120690 | 347 credits completed*

- **14 distinctions across three years of study,** including a 97% in Linear Algebra with Applications.
- **Year 1 (2024), 1 distinction:** Procedural Programming (89%).
- **Year 2 (2025), 6 distinctions:** IT Project Management (89%), Web Server Management (88%), Machine Learning Algorithms (85%), Mathematics 1B (81%), Artificial Intelligence Techniques (77%), Probability and Statistics (77%).
- **Year 3 (2026, in progress), 7 distinctions:** Linear Algebra with Applications (97%), Operating Systems (88%), 4IR Technologies (84%), Sensors Data Management, IoT Mining and Analytics (82%), Object Oriented Systems Analysis and Design (80%), R Programming for Data Science (80%), Research Design and Methodology (76%).
- **Further coursework:** Database Systems, Scientific Computing in Python, Data Structures and Algorithms in Python, Network Security, Cloud-Based Technologies, Computer Network Technologies, AI Ethics and Privacy, Business Management and Entrepreneurship, Project: Data Science.

## EXPERIENCE & LEADERSHIP

**Delegate, Youth 20 (Y20), G20** Mar 2025 - Nov 2025

- **Selected to represent youth perspectives** under the G20 framework, contributing to discussions on global economic and social policy.
- **Advanced positions on youth empowerment, innovation and financial inclusion** alongside delegates from member economies.

**Participant, All4Youth Sub-Saharan Africa Leadership Programme** Current

- **Leadership development programme** focused on youth empowerment and professional growth across Sub-Saharan Africa, with training in teamwork, communication, problem-solving and community impact.

**Retail Assistant, Sportscene, Sandton City** Oct 2024 - Apr 2025

- **Served 50+ customers daily** in a high-traffic retail environment while completing full-time studies.
- **Managed inventory, stock control and visual merchandising** to improve product presentation and availability.

## PROFESSIONAL CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate
- Microsoft Certified: Azure Data Scientist Associate
- Microsoft Certified: Implementing Data Engineering Solutions Using Microsoft Fabric
- Kubernetes and Cloud Native Essentials

*References available on request*